

Mini Project 1: Sales, Customer & Order Management Analysis

Formatting & Setup Instructions (Sales_Data)

Step-by-Step Execution:

1. **Font & Typography:** Select all cells (Ctrl + A) and set Font to **Segoe UI**, Size **11pt**.
2. **Header Styling:**
 - Highlight Row 1 (A1:V1).
 - Fill Color: **Dark Navy** (#1B365D). Font Color: **Bold White** (FFFFFF).
 - Row Height: **28pt**. Text Alignment: **Center, Middle Align, Wrap Text**.
3. **Borders & Gridlines:** Apply **Thin Light Grey Borders** (#D2D6DC) across A1:V101. Ensure Gridlines remain enabled (View → Show → Gridlines).
4. **Column Alignment:**
 - **Center Align:** Order_ID, Order_Date, Gender, City, State, Payment_Mode, Order_Status, Delivery_Date, Customer_Type.
 - **Right Align:** Age, Quantity, Unit_Price, Discount, Rating, Total_Amount, Discount_Amount, Net_Sales.
 - **Left Align:** Customer_Name, Product_Category, Product, Salesperson, Remarks.
5. **Freeze Panes:** Select Cell c2, then navigate to View → Freeze Panes → Freeze Panes. This keeps Order_ID, Order_Date, and the Header Row visible while scrolling.

Number Formatting & Custom Formatting Guide (Sales_Data)

1. Indian Currency (Unit_Price, Total_Amount, Net_Sales)

- **Objective:** Display Indian Rupee symbol with comma separation.
- **Steps:** Select range → Ctrl + 1 (Format Cells) → Custom → Enter:
[₹-en-IN] #,##0.00

2. Custom Unit Quantities (Quantity)

- **Objective:** Append the text "Units" while maintaining numeric properties for mathematical operations.
- **Custom Code:**
#,##0" Units"

3. Rating Display (Rating)

- **Objective:** Append a star symbol to rating numbers.
- **Custom Code:**
0.0" ★"

4. Custom Order ID (Order_ID)

- **Objective:** Automatically prefix numbers with ORD- and pad with zeros up to 4 digits.
- **Custom Code:**
"ORD-"0000

Series & AutoFill (Series_Practice)

Exercise	Starting Input	Menu Path / Technique	Resulting Series
Linear Numbers	1, 2	Select both → Drag Fill Handle down	1, 2, 3, 4, 5...
Even Series	2, 4	Home → Fill → Series → Step: 2	2, 4, 6, 8, 10...
Date Series	01-Sep-2026	Drag Fill Handle → Select Fill Days	01-Sep-2026, 02-Sep-2026...
Weekday Series	01-Sep-2026	Drag Fill Handle → Choose Fill Weekdays	Excludes Saturdays & Sundays
Custom List	Beginner	File → Options → Advanced → Edit Custom Lists	Beginner, Intermediate, Advanced, Expert

Interactive Order Entry Form (Data_Entry)

Data Validation Setup Code Reference:

- **Gender List:** Home → Data Validation → List → Source: Male, Female
- **Quantity Range:** Data Validation → Allow: Whole number → Data: between → Minimum: 1, Maximum: 100
- **Error Alert Tab:** Set Title: Invalid Entry, Message: Please select a value from the allowed list.

Mathematical & Conditional Formulas (Analysis)

1. Overall Mathematical Metrics

Metric	Business Problem	Excel Formula	Expected Output
Total Quantity Sold	How many total items were sold across all orders?	=SUM(Sales_Data!K2:K101)	Dynamic Count
Total Gross Revenue	What is the total revenue before discounts?	=SUM(Sales_Data!T2:T101)	₹ Sum
Total Net Sales	What is the total realized revenue after discounts?	=SUM(Sales_Data!V2:V101)	₹ Sum
Average Unit Price	What is the average price of products sold?	=AVERAGE(Sales_Data!L2:L101)	₹ Avg
Highest Unit Price	What is the maximum product price?	=MAX(Sales_Data!L2:L101)	₹ Max
Lowest Unit Price	What is the minimum product price?	=MIN(Sales_Data!L2:L101)	₹ Min
Average Discount %	What is the store-wide average discount rate?	=AVERAGE(Sales_Data!M2:M101)	Percentage
Total Order Volume	How many valid order records exist?	=COUNTA(Sales_Data!A2:A101)	100

2. Statistical Counting (COUNT, COUNTA, COUNTIF)

- Count Completed Orders: =COUNTIF(Sales_Data! O2:O101, "Completed")
- Count Pending Orders: =COUNTIF(Sales_Data! O2:O101, "Pending")
- Count Cancelled Orders: =COUNTIF(Sales_Data! O2:O101, "Cancelled")
- Count Orders from Malegaon: =COUNTIF(Sales_Data! F2:F101, "Malegaon")
- Count Electronics Orders: =COUNTIF(Sales_Data! H2:H101, "Electronics")
- Count 5-Star Rated Orders: =COUNTIF(Sales_Data! Q2:Q101, 5)

3. Conditional Aggregations (SUMIF & AVERAGEIF)

- Total Revenue from Electronics Category:
=SUMIF(Sales_Data!H2:H101, "Electronics", Sales_Data!V2:V101)
- Total Revenue Generated in Malegaon City:
=SUMIF(Sales_Data!F2:F101, "Malegaon", Sales_Data!V2:V101)
- Total Revenue Generated by Salesperson "Amit Sharma":
=SUMIF(Sales_Data!J2:J101, "Amit Sharma", Sales_Data!V2:V101)
- Average Customer Rating for Office Supplies:
=AVERAGEIF(Sales_Data!H2:H101, "Office Supplies", Sales_Data!Q2:Q101)
- Average Net Sales Value per Completed Order:
=AVERAGEIF(Sales_Data!O2:O101, "Completed", Sales_Data!V2:V101)

4. Practical Business Question Answers

1. Average Order Value (AOV): =AVERAGE(Sales_Data!V2:V101)
2. New vs. Returning Customer Count:
=COUNTIF(Sales_Data!R2:R101, "New")
=COUNTIF(Sales_Data!R2:R101, "Returning")
3. Average Delivery Time across all Orders: =AVERAGE(Sales_Data!P2:P101 - Sales_Data!B2:B101)

Date & Time Arithmetic (Date_Practice)

- Extract Order Year: =YEAR(Sales_Data!B2)
- Extract Order Month Number: =MONTH(Sales_Data!B2)
- Extract Day of Month: =DAY(Sales_Data!B2)
- Extract Full Day Name (e.g., "Monday"): =TEXT(Sales_Data!B2, "dddd")
- Extract ISO Week Number: =WEEKNUM(Sales_Data!B2)
- Calculate Delivery Lead Duration (in Days): =Sales_Data!P2 - Sales_Data!B2
- Calculate Month-End Expiration Date: =EOMONTH(Sales_Data!B2, 0)

Data Cleaning & Text Functions (Text_Practice)

- Clean Extra Spaces & Apply Proper Title Casing:
=PROPER(TRIM(Sales_Data!C2))
- Count Character Length of Customer Name:
=LEN(B2)
- Extract Order Code Prefix (First 3 Characters):
=LEFT(Sales_Data!A2, 3) --> Result: "ORD"
- Extract Numeric Order Sequence (Last 4 Characters):
=RIGHT(Sales_Data!A2, 4) --> Result: "1001"
- Combine City and State into Full Address String:
=TEXTJOIN(" ", TRUE, Sales_Data!F2, Sales_Data!G2) --> Result: "Malegaon, Maharashtra"

Advanced & Multi-Level Sorting (Sales_Data)

1. Multi-Level Custom Business Sort:

Sort the dataset to prioritize regional executive hierarchies:

- **Level 1:** State (A to Z)
- **Level 2:** City (A to Z)
- **Level 3:** Product_Category (A to Z)
- **Level 4:** Net_Sales (Largest to Smallest)

2. Custom List Sort (Order_Status Priority):

To display operational status in logical order (**Completed** → **Pending** → **Cancelled**):

1. Navigate to **Data** → **Sort** → **Order_Status**.
2. Under **Order**, select **Custom List...**
3. Type: **Completed, Pending, Cancelled** → Click **Add** → Click **OK**.

Filtering & Conditional Formatting Rules (Sales_Data)

Conditional Formatting Setup:

1. **High Sales Highlight:**
 - **Range:** Net_Sales (V2:V101)
 - **Rule:** Cell Value > ₹20,000 → Fill **Soft Green** (#E6F4EA), Text **Dark Green** (#137333).
2. **Data Bars on Sales:**
 - **Range:** Total_Amount (T2:T101)
 - **Type:** Gradient Fill **Blue Data Bars**.
3. **Rating Color Scale:**
 - **Range:** Rating (Q2:Q101)
 - **Type:** 3-Color Scale (Red = 1.0, Yellow = 3.0, Green = 5.0).
4. **Order Status Rules:**
 - Completed → Light Green Fill
 - Pending → Soft Yellow Fill
 - Cancelled → Soft Red Fill
5. **Duplicate Order ID Detection:**

- Select A2:A101 → **Home** → **Conditional Formatting** → **Highlight Cells Rules** → **Duplicate Values...**

Dashboard Layout & Final Practical Examination

Dashboard Sheet Highlights:

- **8 KPI Summary Cards:** Live tracking of Orders, Gross Revenue, Net Revenue, Quantity, Average Order Value, Completed/Pending/Cancelled counts, and Average Rating.
- **Embedded Charts:**
 1. **Column Chart:** Revenue breakdown by Product Category.
 2. **Pie / Doughnut Chart:** Fulfillment Status proportions (Completed vs. Pending vs. Cancelled).

Students should complete these 10 tasks in the Final Challenge tab to test mastery:

#	Task	Excel Formula / Solution Method
1	Top 10 Orders by Net Sales	Data → Sort by <u>Net_Sales</u> (Descending) OR =LARGE(Sales_Data!V2:V101, {1,2,3...})
2	Top 5 Products by Quantity	=SUMIF(Sales_Data!I2:I101, "Laptop", Sales_Data!K2:K101) evaluated across unique items
3	Top Revenue City	Compute =SUMIF() for each city; identify max using =MAX()
4	Monthly Sales Calculation	Combine =MONTH() column with =SUMIF() grouping
5	Top Salesperson	Evaluate =SUMIF(Sales_Data!J2:J101, Salesperson_Name, Sales_Data!V2:V101)
6	Filter High-Value Loyal Customers	Apply Advanced Filter: Customer_Type = "Returning", Rating >= 4, Net_Sales > 10000
7	Delayed Pending Orders	=IF(AND(Sales_Data!O2="Pending", Sales_Data!P2 < TODAY()), "OVERDUE", "On Time")
8	Status Order Sort	Custom List Sort: Completed → Pending → Cancelled
9	Dual-Condition Formatting	Custom Formula Rule: =AND(V2 > 20000, Q2 >= 4)

10	Executive Presentation	Verification of clear formatting and error-free formulas
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